NM2 - cv3

Ivan Kubiš

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$$
\begin{aligned}
\frac{dS(t)}{dt} &= n(1-f)N - \frac{bI(t)S(t)}{N} - mS(t) \\
\frac{dI(t)}{dt} &= \frac{bI(t)S(t)}{N} - cI(t) - mI(t) \\
\frac{dR(t)}{dt} &= cI(t) + nfN - mR(t)
\end{aligned}
\tag{1}
$$

- N: celková velikost populace,
- n: míra porodnosti,
- m: míra přirozené úmrtnosti ($m \approx n$),
- b: koeficient rizika nakažení (transmise),
- c: míra uzdravení (recovery rate),
- f: podíl očkované populace (proočkovanost),
- $\frac{bI(t)S(t)}{N}$: počet nově nakažených jedinců za jednotku času.

Parametry úlohy:

1. Čas 0 - 200
2. $n = 0.1$
3. $f = 0.2$
4. $b = 0.5$
5. $c = 0.3$
6. $m = 0.06$
7. $mi = 0.065$

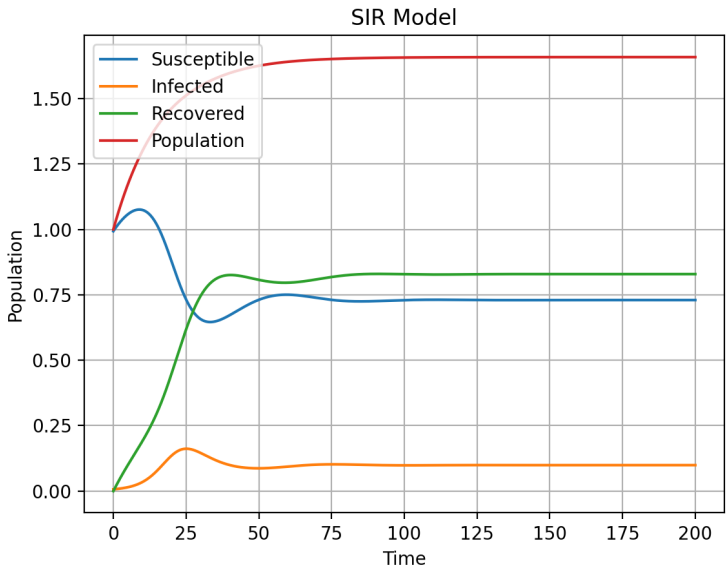

Figure 1: $n = 0.1$

1 Birth rate

Je vidět že hlavní vliv parametru porodnosti je na velikost populace, zajímavé je, že se počet infikovaných ustálí v téměř vždy stejný čas nezávisle na tomto parametru.

2 Proočkovanost

Zde je vidět, že proočkovanost má vliv hlavně na počet infikovaných a uzdravených lidí. Je zajímavé, že pro f = 0.6 infekce dokonce zcela vymizí.

3 Přirozená mortalita

V této části byl měněn parametr přirozené mortality. Je vidět, že tento parametr ovlivňuje všechny hodnoty, ale hlavně počet populace.

4 Koeficient přenosu

Zde byl měněn parametr přenosu b. Má vliv hlavně na počet uzdravených a odolných lidí.

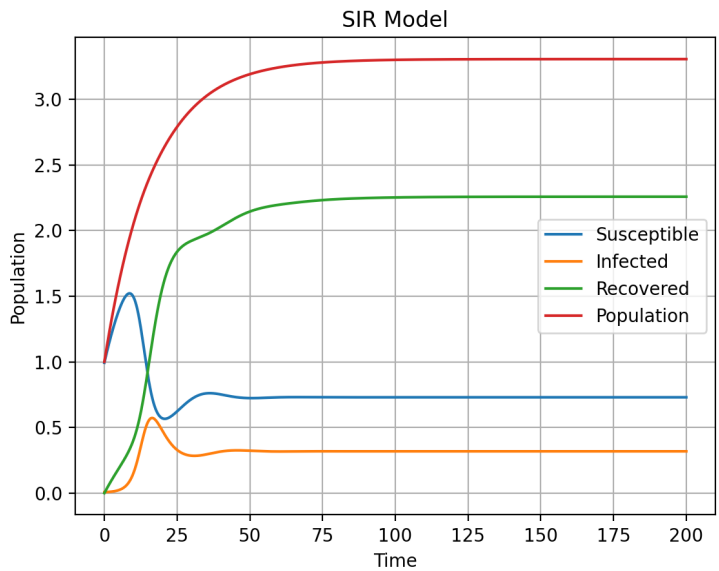

Figure 2: $n = 0.2$

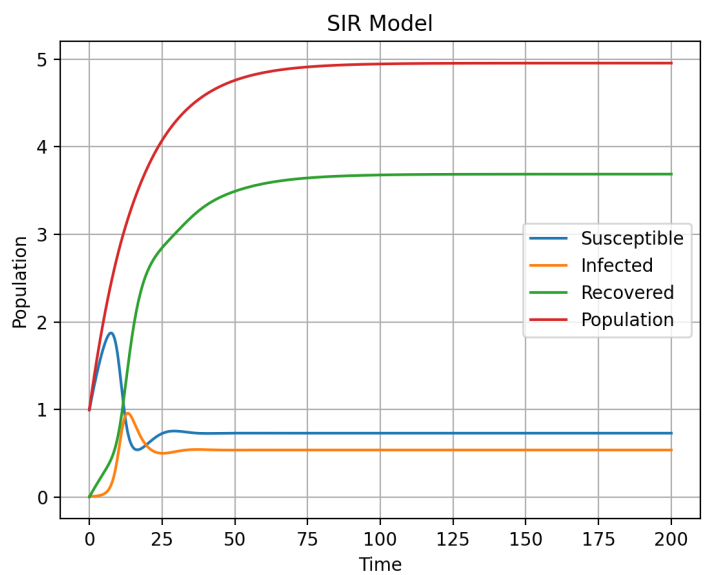

Figure 3: $n = 0.3$

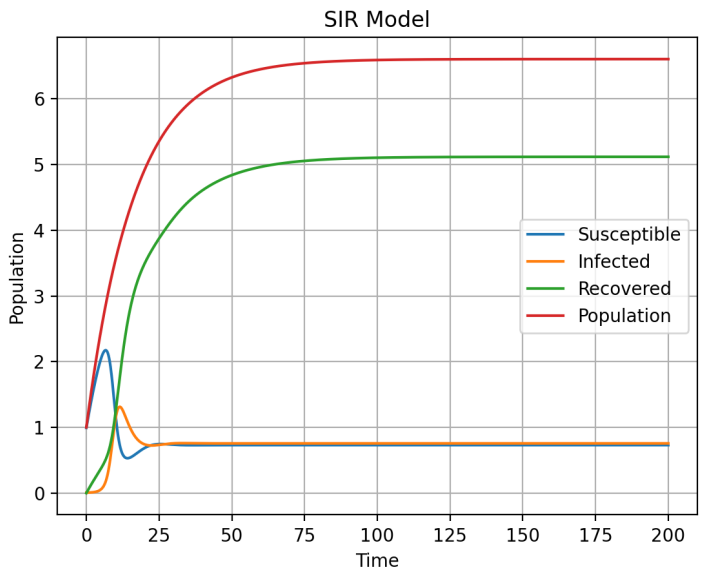

Figure 4: $n = 0.4$

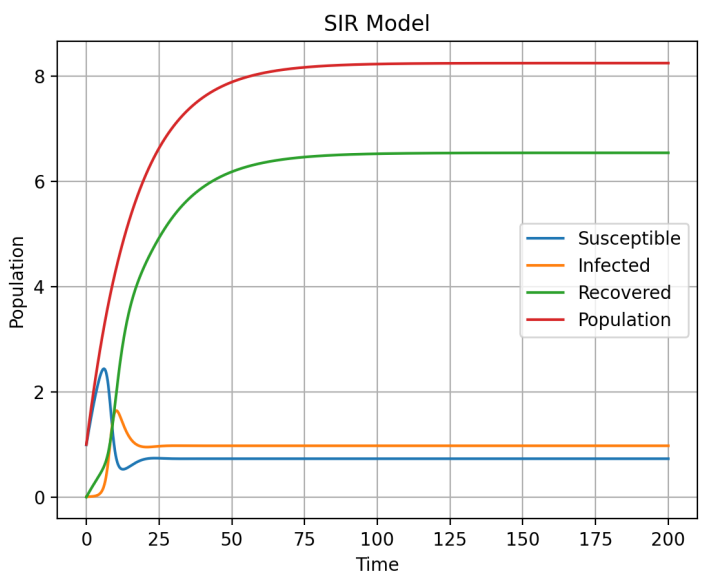

Figure 5: $n = 0.5$

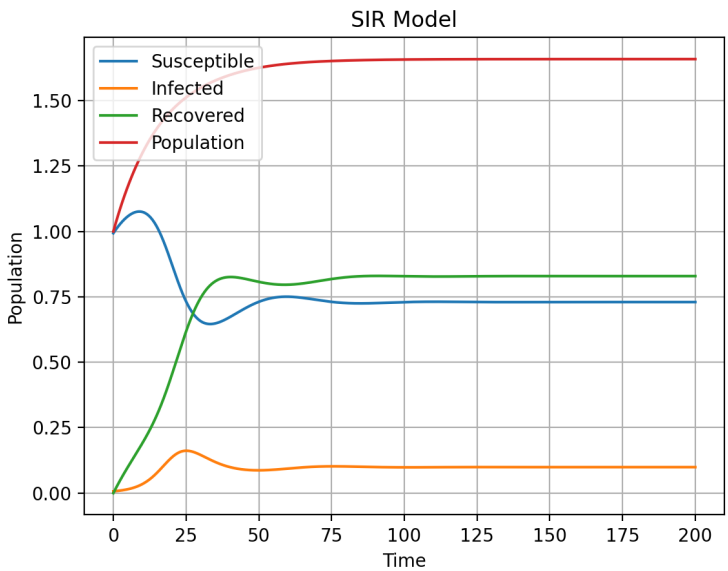

Figure 6: $f = 0.2$

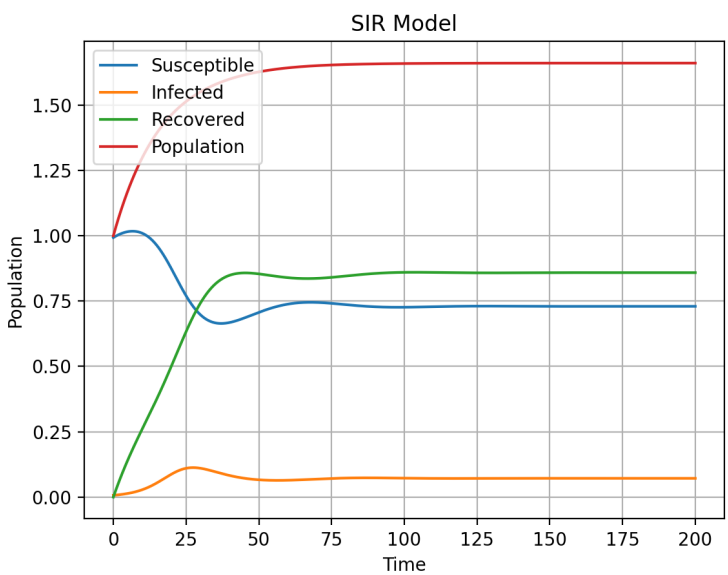

Figure 7: $f = 0.3$

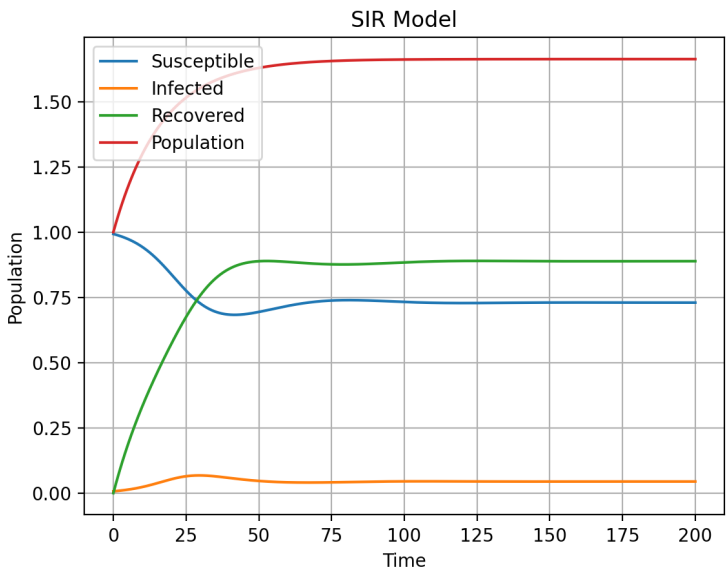

Figure 8: $f = 0.4$

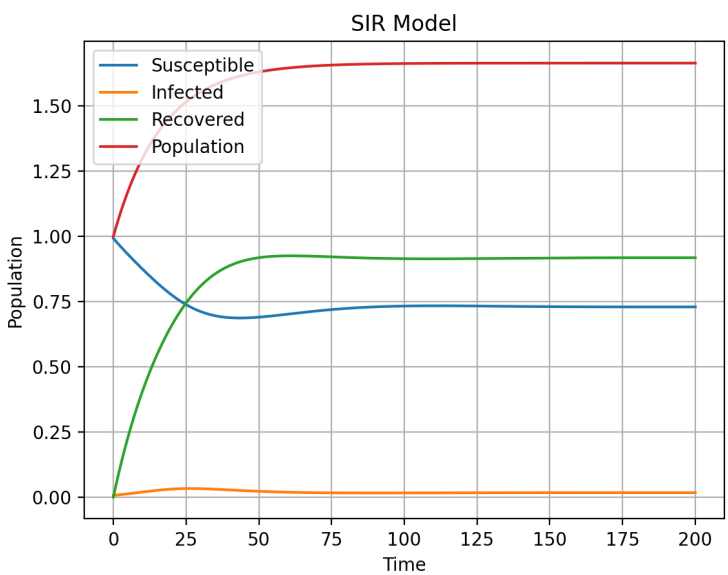

Figure 9: $f = 0.5$

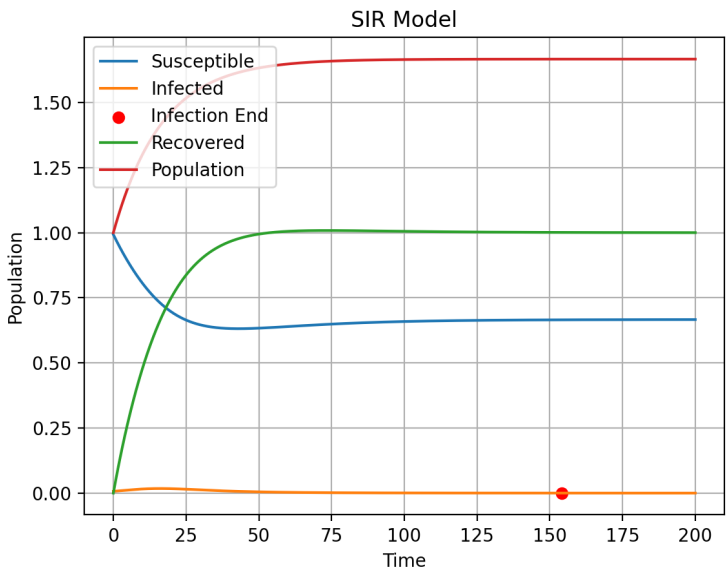

Figure 10: $f = 0.6$

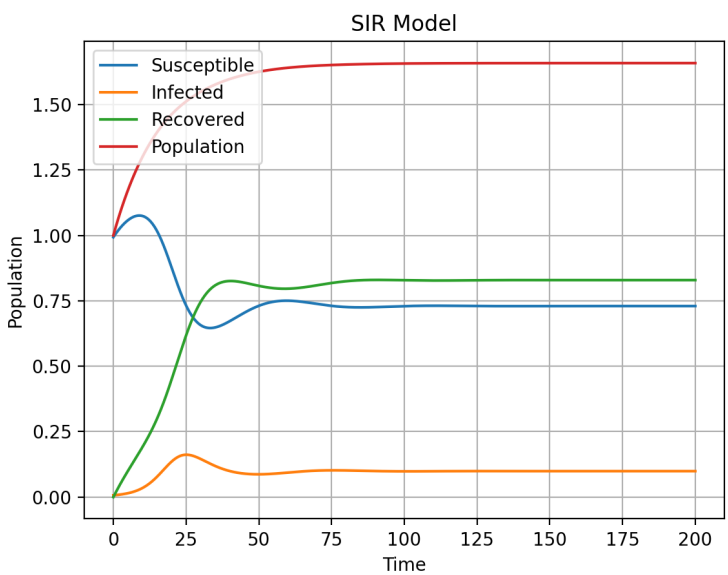

Figure 11: $m = 0.06$

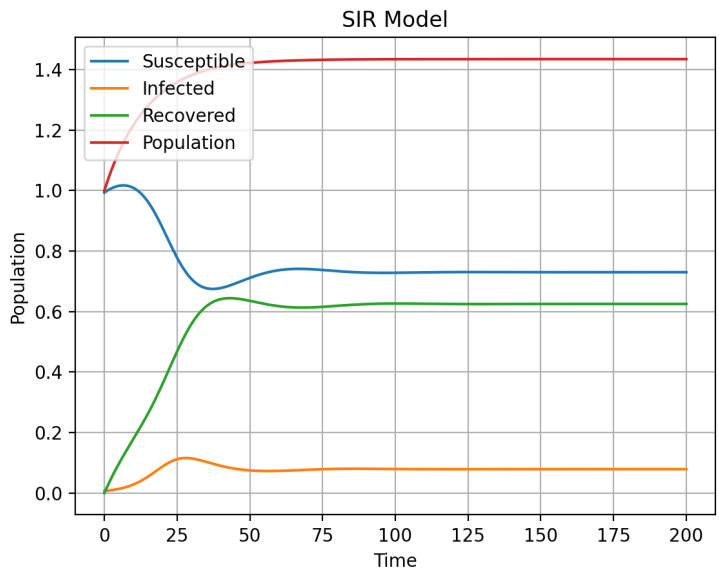

Figure 12: $m = 0.07$

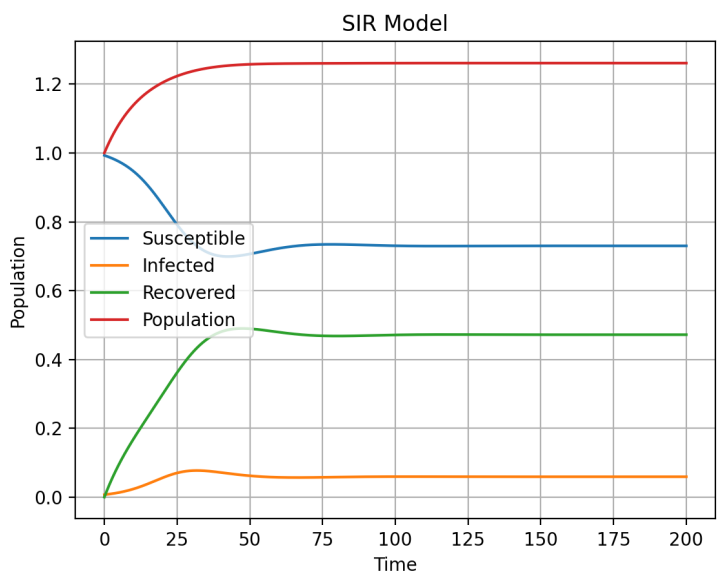

Figure 13: $m = 0.08$

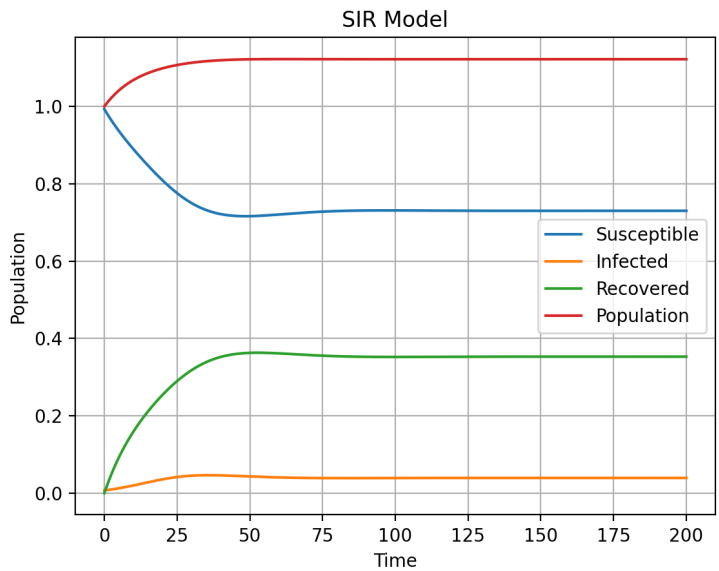

Figure 14: $m = 0.09$

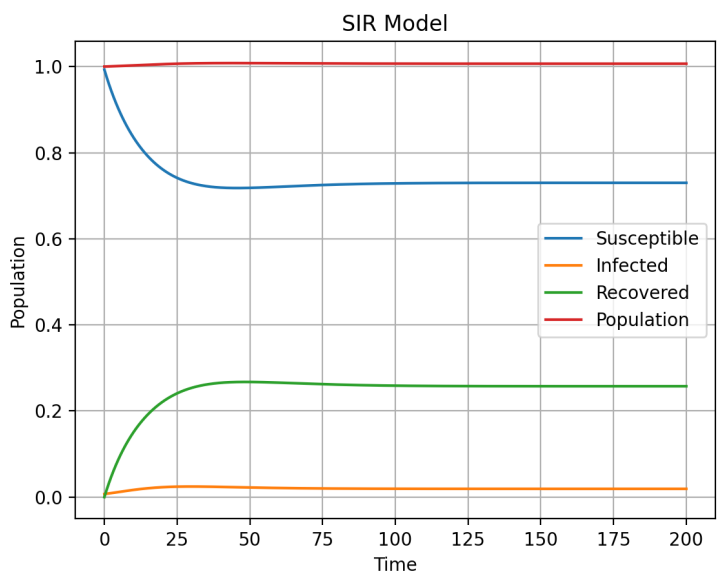

Figure 15: $m = 0.1$

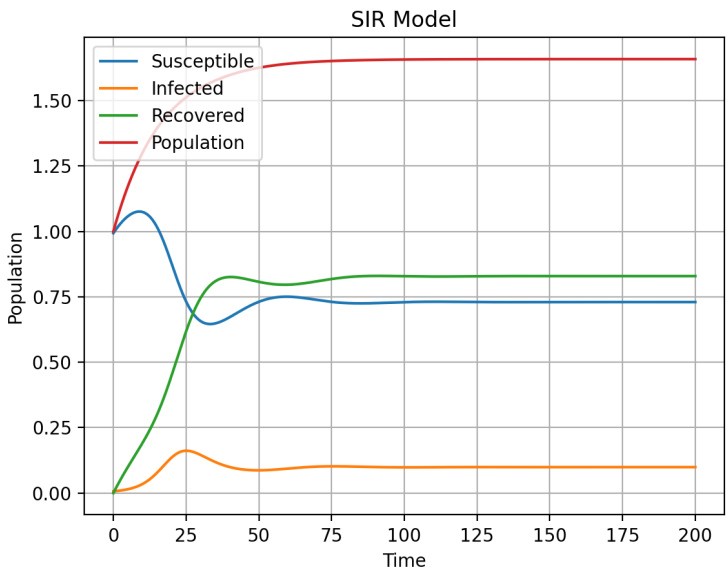

Figure 16: $b = 0.5$

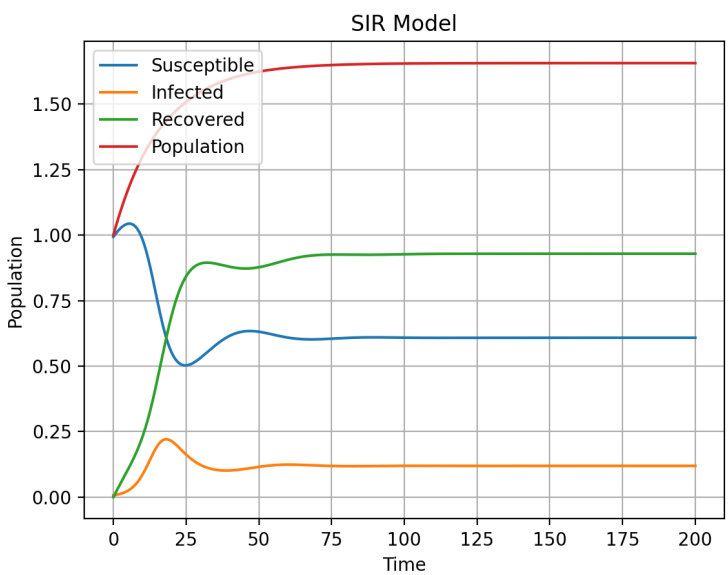

Figure 17: $b = 0.6$

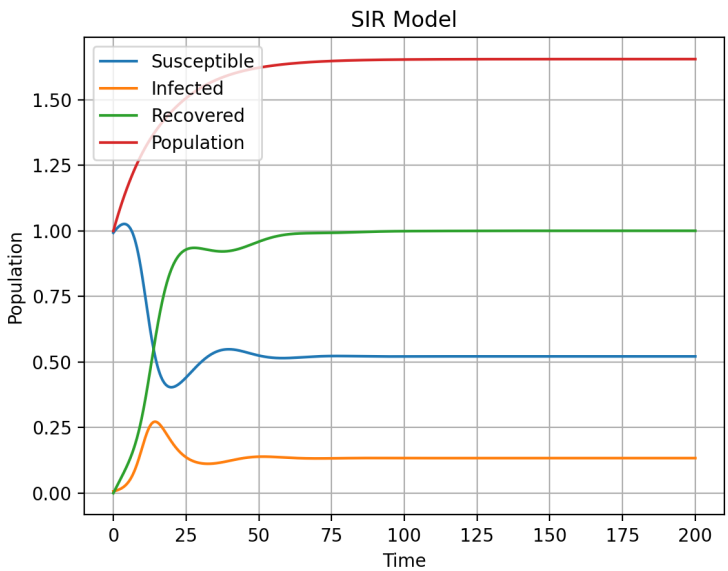

Figure 18: $b = 0.7$

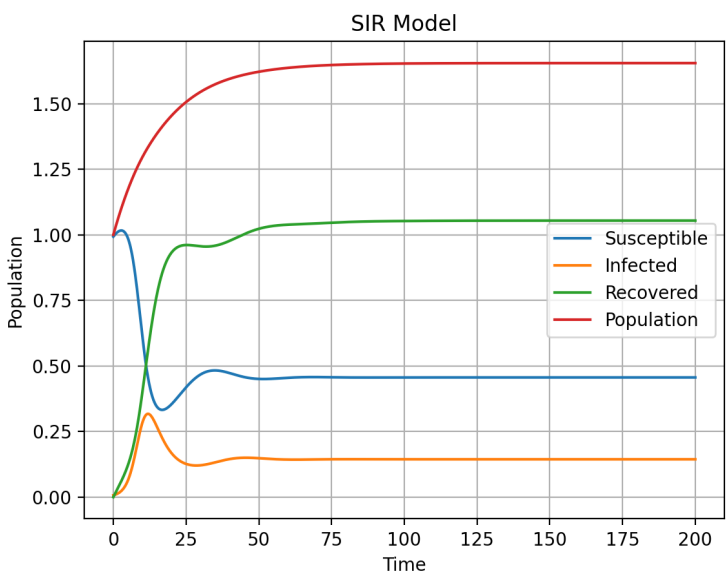

Figure 19: $b = 0.8$

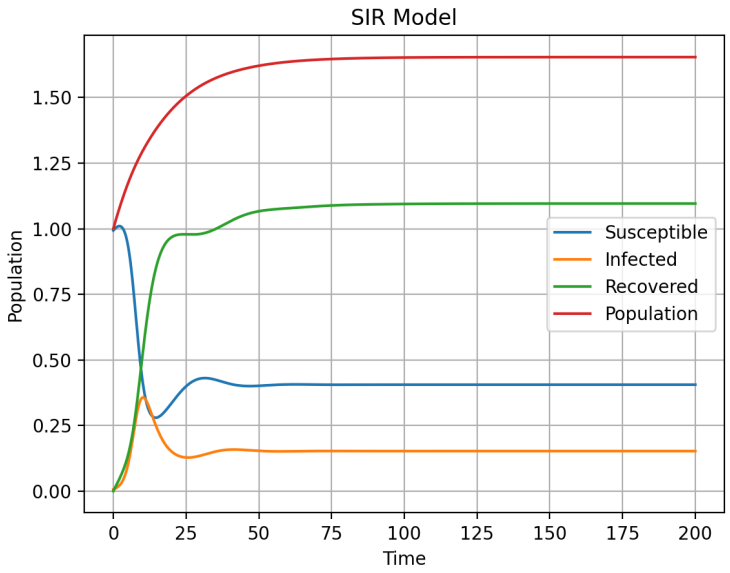

Figure 20: $b = 0.9$

5 Reálná situace

Zde byly vzaty data z dob covidu v ČR, je vidět že šlo zejména o rychlý nárůst počtu infikovaných.

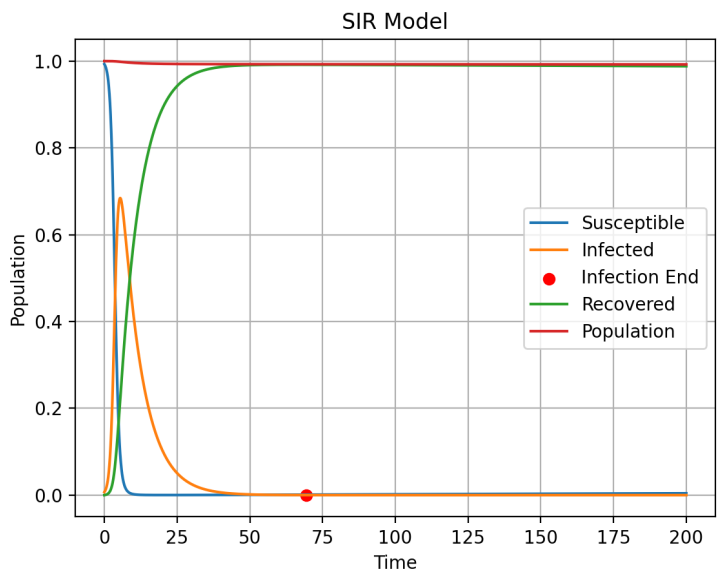

Figure 21: $n = 0.000028$, $f = 0.15$, $b = 1.5$, $c = 0.14$, $m = 0.000032$, $mi = 0.001$